Data Warehousing and Modeling

Lecture 2
Introduction to Data warehousing

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What is a Data Warehouse?

- A **Data Warehouse** plays a central role in Business Intelligence systems by collecting, integrating and analyzing data from various data sources.
- **Data Warehousing** is a **data integration** environment that combines technologies and components to store, query and analyze large volumes of data, transforming it into valuable and accessible information for users.
- Unlike a company's operational database, a data warehouse allows access to historical and current data, thus facilitating informed decision making.
- **Data Warehousing** also refers to the process of collecting and managing data to extract valuable information.

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Data Warehouse is a centralized repository that stores, manages, and consolidates data from various sources within an organization. It is designed to support business intelligence (BI) activities, reporting, and data analysis.

States of a Data Warehouse

A data warehouse can be in two states:

Data Warehouse Offline: Data is copied from one operating system to another server, where loading, processing and reporting do not affect the performance of the operating system.

Data Warehouse Online: Data is regularly updated from the operational database. In the case of a real-time data warehouse, the update occurs every time a transaction occurs in the relational database, such as in train or aircraft reservation systems.

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Historical Perspective: Inmon and Kimball

.1. Bill Inmon - The Father of Data

In **1970, Nielsen and IRI** introduced the concept of **dimensional data marts** for retail, and in **1983,** Teradata launched a database management system designed specifically for decision support.

In the 1980s, Inmon introduced the idea of the **"Corporate Information Factory" (CIF),** which laid the foundation for modern data warehousing. The CIF concept emphasized the centralization of data into a single, integrated repository. Inmon's approach focused on creating a normalized data model, known as the **"Enterprise Data Warehouse (EDW)".**

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Historical Perspective: Inmon and Kimball

2. Ralph Kimball - The Dimensional Modeling Pioneer

Ralph Kimball is renowned for his pioneering work in dimensional modeling, which is a key component of data warehousing.

In the 1990s, Kimball introduced the concept of the "Data Warehouse Bus Architecture." Unlike Inmon's approach, Kimball's methodology focused on creating data marts, which are smaller, subject-specific data warehouses derived from the central data warehouse.

Kimball's dimensional modeling emphasized the use of fact tables and dimension tables, which simplified data retrieval and analysis. This approach made it easier for business users to understand and navigate data.

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Challenges of Traditional Data Management

- **Data Fragmentation**: In traditional setups, data is often scattered across various systems, departments, and formats, creating data silos.

- **Limited Historical Data**: Operational databases, known as OLTP systems, are designed for transaction processing. They typically store a limited history of data, making it challenging to perform historical trend analysis and long-term forecasting.

- **Inefficient Reporting**: Generating meaningful reports from multiple, disconnected data sources is time-consuming and error-prone.
- **Resource-Intensive Queries:** Analyzing data from operational databases can overload these systems, leading to slower performance and potentially disrupting day-to-day operations.

- **Lack of Integration:** Integrating data from different sources and formats is complex and often requires custom scripts or manual efforts.

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Importance of Data Warehousing

- ✓ **Data Integration**: Unifies diverse data sources, revealing insights across departments.
- ✓ **Historical Analysis**: Provides a long-term view for trend analysis and informed decision-making.
- ✓ **Enhanced BI**: Empowers data-driven decisions through robust reporting and BI tools.
- ✓ **Efficient Queries**: Optimized for quick, complex data analysis without affecting transactional systems.
- ✓ **Competitive Edge:** Enables agility, responsiveness, and innovation for a competitive advantage.
- ✓ **Compliance & Governance:** Ensures data accuracy, consistency, and security, crucial in regulated industries.
- ✓ **Scalability:** Adapts to growing data needs, suitable for startups to large enterprises.
- ✓ **Cost Savings:** Reduces storage costs, streamlines data management, and minimizes duplication.
- ✓ **Data-Driven Culture:** Cultivates a culture of empirical decision-making, transforming business operations.

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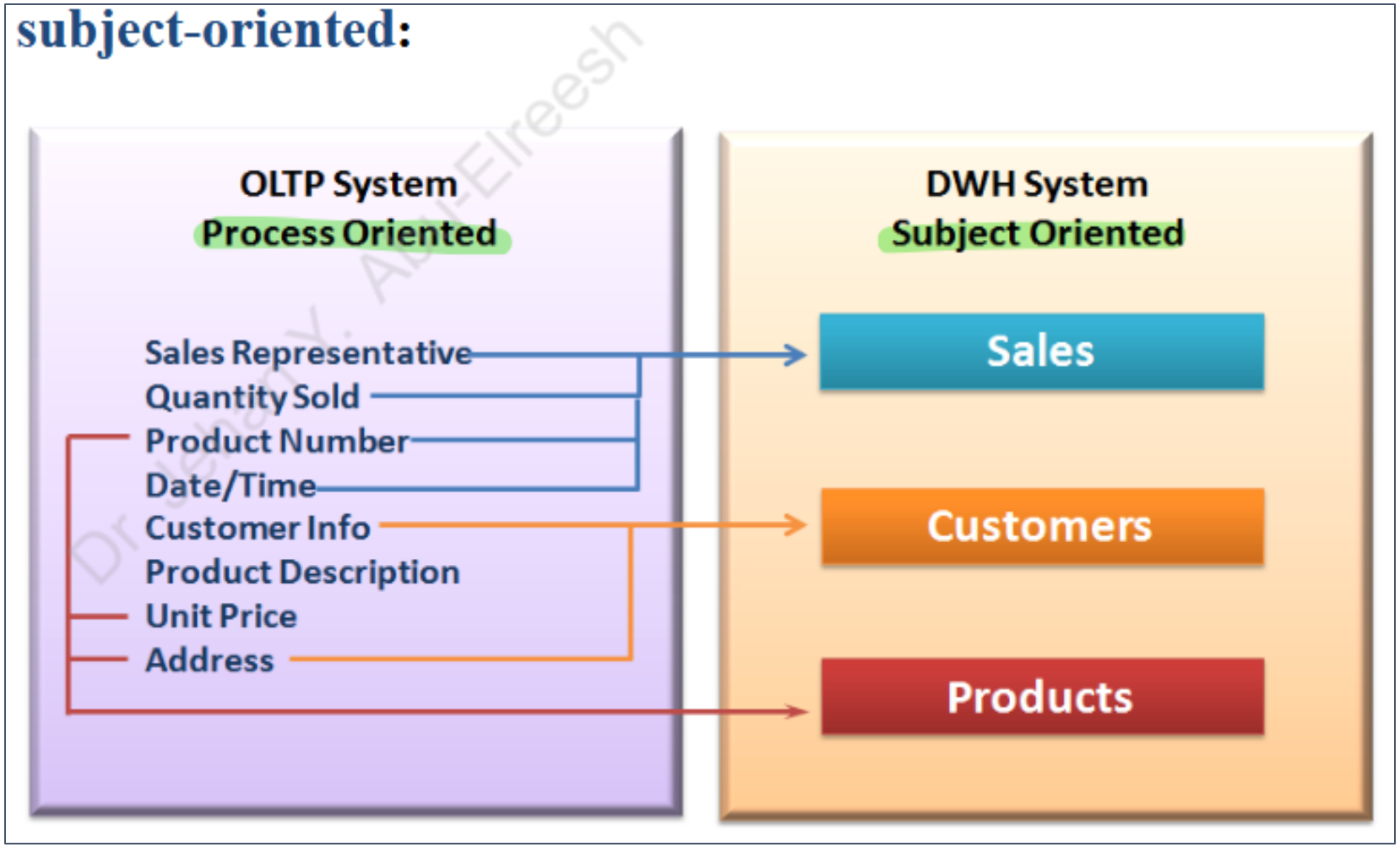
Data warehouse characteristics

1- Subject-Oriented

Provide **a simple and concise** view around particular subject issues by **excluding data that are not useful in the decision support process**



Data warehouse characteristics

2- Integrated

Constructed by integrating multiple, heterogeneous data sources, relational databases, flat files, on-line transaction records

3- Time Variant

The time horizon for the data warehouse is significantly longer than that of operational systems

- Operational database: current value data
- Data warehouse data: provide information from a historical perspective (e.g., past 5-10 years)

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Data warehouse characteristics

4- Nonvolatile

- **A physically separate store** of data transformed from the operational environment
- Operational **update of data does not occur** in the data warehouse environment
  - Does not require transaction processing, recovery, and concurrency control mechanisms
  - Requires only two operations in data accessing:
    - *initial loading of data* and **access of data.**

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Types of Data Warehousing

There are three main types of data warehouse:

1- Enterprise Data Warehouse (EDW)

This type of warehouse serves as a key or central database that facilitates decision-support services throughout the enterprise. The advantage to this type of warehouse is that it provides access to cross-organizational information, offers a unified approach to data representation, and allows running complex queries.

2- Operational Data Store (ODS)

This type of data warehouse refreshes in real-time. It is often preferred for routine activities like storing employee records. It is required when data warehouse systems do not support reporting needs of the business.

3- Data Mart

A data mart is a subset of a data warehouse built to maintain a particular department, region, or business unit.

Every department of a business has a central repository or data mart to store data.

The data from the data mart is stored in the ODS periodically. The ODS then sends the data to the EDW,

where it is stored and used.

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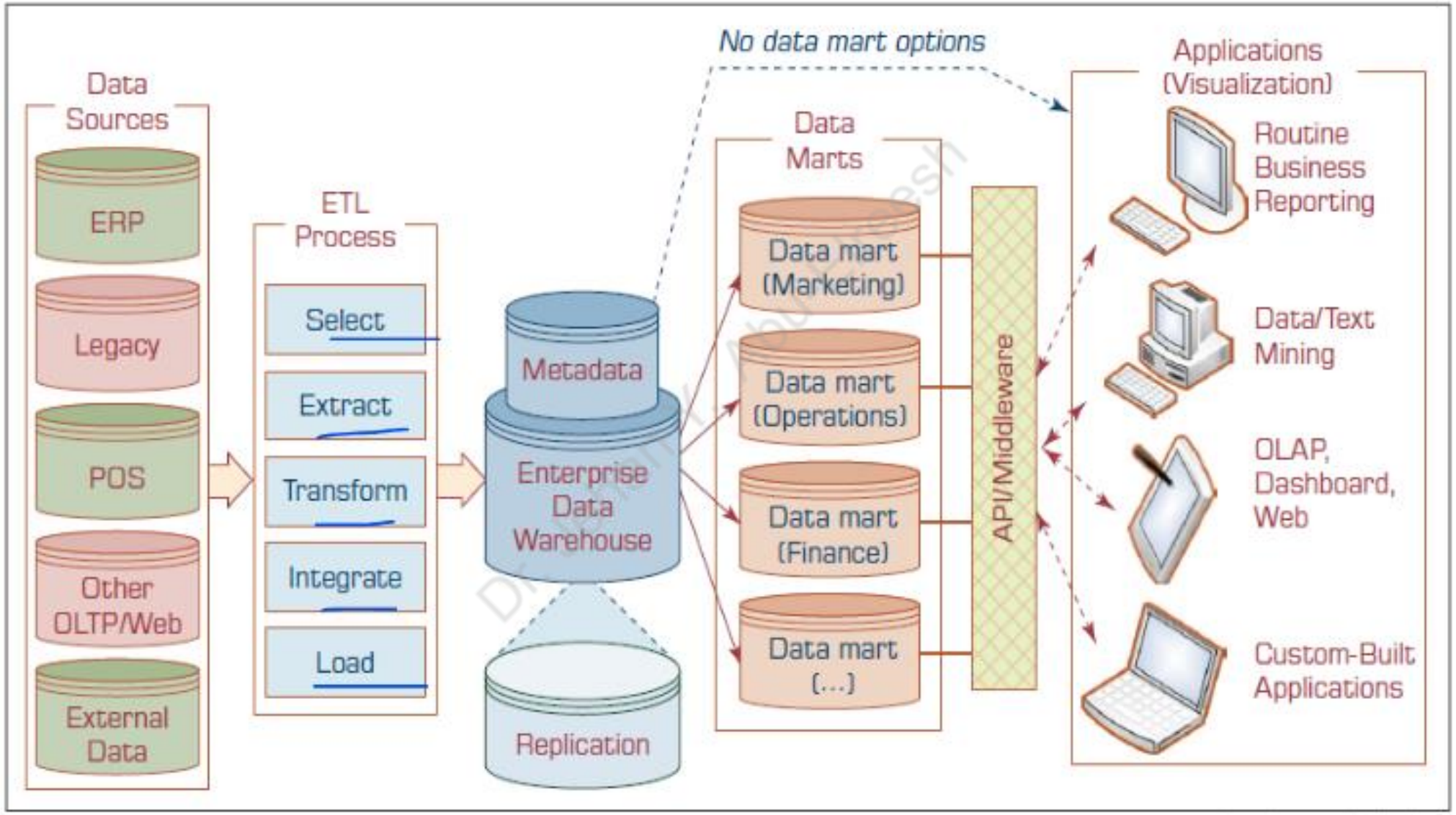
Data Warehouse Framework



Metadata Repository

Meta data is the data defining warehouse objects. It stores:

- Description of the **structure** of the data warehouse
  - schema, view, dimensions, hierarchies, derived data defining, data mart locations and contents
- **Operational** meta-data
  - data lineage (history of migrated data and transformation path), currency of data (active, archived, or purged), monitoring information (warehouse usage statistics, error reports, audit trails)
- The **algorithms** used for summarization
- The **mapping** from operational environment to the data warehouse
- Data related to **system performance**
  - warehouse schema, view and derived data definitions
- **Business data**
  - business terms and definitions, ownership of data, charging policies

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How Does a Data Warehouse Work?

- A **data warehouse** works as the **central data repository** for an organization, acting as a convergence point for information from various sources.
- This process involves the extraction of data from its original sources, followed by integration into the data warehouse. The data is then processed, transformed and organized into **views, dimension tables and fact tables**, using methodologies such as **ETL (Extract, Transform and Load)** or ELT (Extract, Load and Transform).
- Once the data has been transformed and organized, users can access it in a variety of ways, whether through **SQL queries**, business intelligence tools such as **Power BI**, or customer management platforms such as a CRM.
- The data warehouse provides a layer of abstraction that facilitates **access to data** in a structured and coherent way, allowing users to obtain relevant information for business decisions.

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How Does a Data Warehouse Work?

- This central repository, the **data warehouse**, is vital for obtaining a complete view of critical elements, such as the customer base.
- By **consolidating information** in one place, a comprehensive review of all data is ensured, also enabling data mining.
- This process involves finding trends and patterns in the data, which are used to drive strategies to increase sales and revenue for the company.
- In short, the data warehouse is an essential tool for effective **data management** and informed business decision-making.

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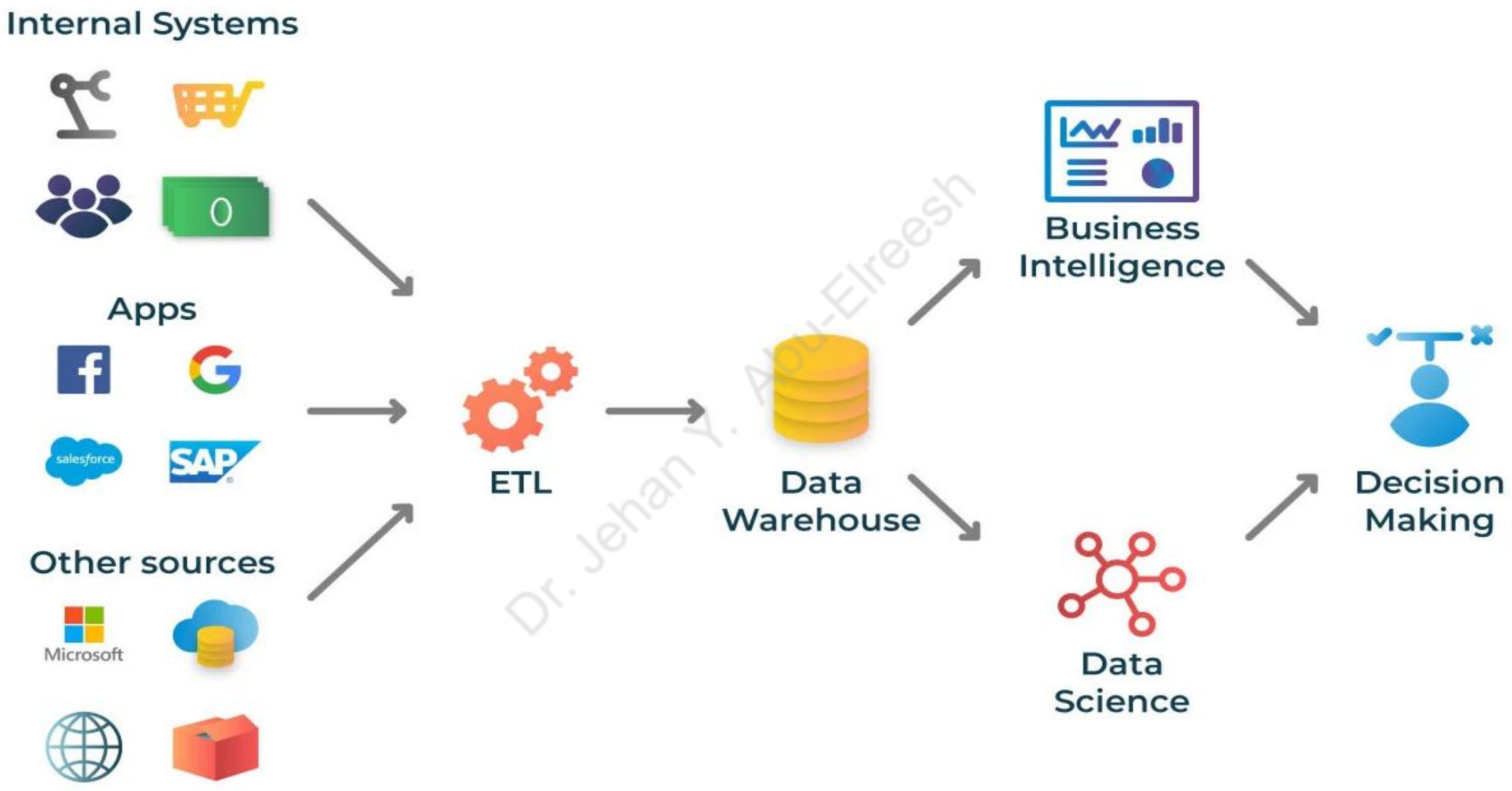

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OLAP vs OLTP

OLTP (Online Transaction Processing)

Primary purpose: Efficiently manage and process day-to-day transactional data.

Characteristics:

- Fast response times.
- Focused on individual transactions.
- Normalized data structure.
- Concurrent user support.

Common use cases:

- Point-of-sale systems.
- Online banking.
- E-commerce order processing.

Key benefits:

- Ensures data integrity.
- Supports high transaction volumes.
- Real-time data updates.

Key technologies:

- Relational databases (e.g., MySQL, Oracle).
- ACID (Atomicity, Consistency, Isolation, Durability) transactions.

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OLAP vs OLTP

OLAP (Online Analytical Processing)

Primary purpose: Perform complex, multidimensional analysis on historical data.

Characteristics:

- Query-intensive.
- Aggregated and summarized data.
- Denormalized data structure (star or snowflake schema).
- Decision support and business intelligence.

Common use cases:

- Business reporting and analytics.
- Data mining and forecasting.
- Market research.

Key benefits:

- Supports complex analytical queries.
- Enables historical and trend analysis.
- Facilitates data slicing, dicing, and drilling.

Key technologies:

- Data warehousing (e.g., Microsoft SQL Server Analysis Services, IBM Cognos).
- OLAP cubes.
- Dimensional modeling.

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OLTP vs. OLAP

CATEGORY	OLTP	OLAP
QUERY TYPES	It process simple queries.	It process complex queries.
CHARACTERISTICS	Manages large amount of small transactions.	Manages large volume of data.
QUERY	Processes Insert, Update and Delete queries.	Processes select queries.
RESPONSE TIME	The Responses time is in miliseconds.	Response time is in seconds or minutes.
DATABASE	Uses traditional database.	Uses Data warehouse.
SOURCE	OLTP is the source of data	Various OLTP databases becomes source of data for OLAP.

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Database vs. Data Warehouse

Although a data warehouse and a traditional database share some similarities, they need not be the same idea.

- The main difference is that in a database, data is collected **for multiple transactional purposes**. However, in a data warehouse, data is collected on an extensive scale to **perform analytics.**

- Databases provide **real-time data**, while warehouses store data to be **accessed for big analytical queries.**

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Why a Separate Data Warehouse?

➢ **High performance for both systems**

- **DBMS— tuned for OLTP:** access methods, indexing, concurrency control, recovery.
- **Warehouse—tuned for OLAP:** complex **OLAP** queries, multidimensional view, consolidation.

➢ **Different functions and different data:**

- **Missing data:** Decision support requires historical data which operational DBs do not typically maintain.
- **data consolidation:** DS requires consolidation (aggregation, summarization) of data from heterogeneous sources
- **data quality:** different sources typically use inconsistent data representations, codes and formats which have to be reconciled

✓ Note: There are more and more systems which perform OLAP analysis directly on relational databases.

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Qustions ??
Thank You

The next Lecture will be :

Introduction to Data Warehouse

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